Data Centers Project

Scalable Architecture for Multi-Departmental Campus

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Course: IT 401 - Data Centers

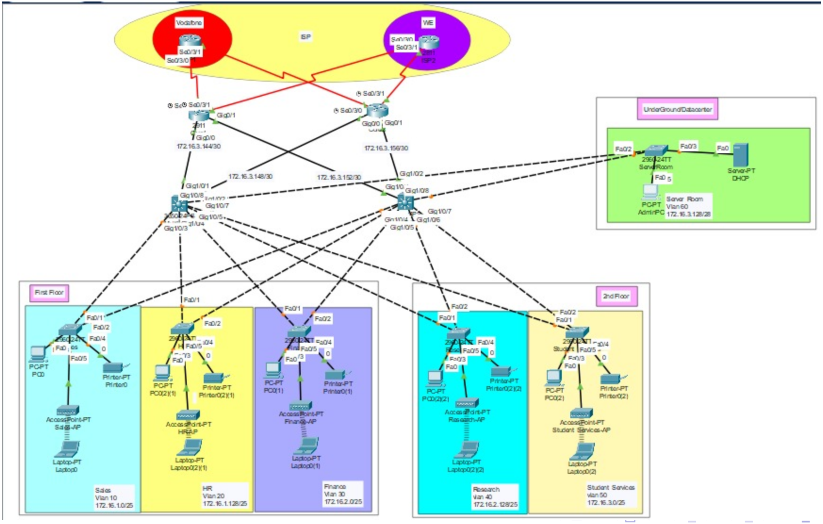
Project Overview

Objective: Design a robust network infrastructure for a 3-story building supporting 600+ users.

- **Departments:** Sales, HR, Finance, Research, Student Services.
- **Critical Assets:** Dedicated Server Room (Underground).
- **Key Requirements:**
  - High Availability (Dual ISP Redundancy).
  - Logical Segmentation (VLANs).
  - Dynamic Routing (OSPF).
  - Centralized Management (SNMP).

Network Topology Design

The design utilizes a hierarchical 3-tier architecture (Core, Distribution, Access) to ensure scalability.



IP Addressing Strategy

We utilized **VLSM** to efficiently allocate the `172.16.1.0/22` block.

Dept	**VLAN**	**Subnet**	**Mask**
Sales	10	172.16.1.0	/25
HR	20	172.16.1.128	/25
Finance	30	172.16.2.0	/25
Research	40	172.16.2.128	/25
Student Svcs	50	172.16.3.0	/25
Server Room	60	172.16.3.128	/28

Table: VLAN & IP Allocation Table

Implementation: VLAN Segmentation

Security through Isolation: Each department resides in a separate broadcast domain.

- VLANs 10-60 Created.
- Inter-VLAN routing via Multilayer Switches (SVIs).

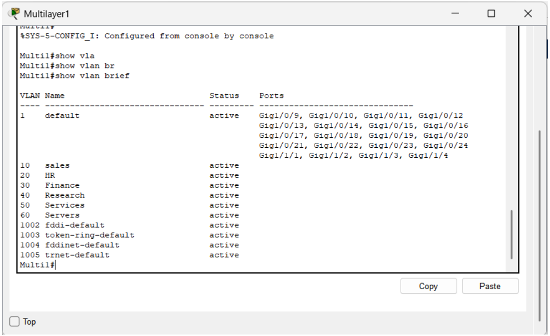
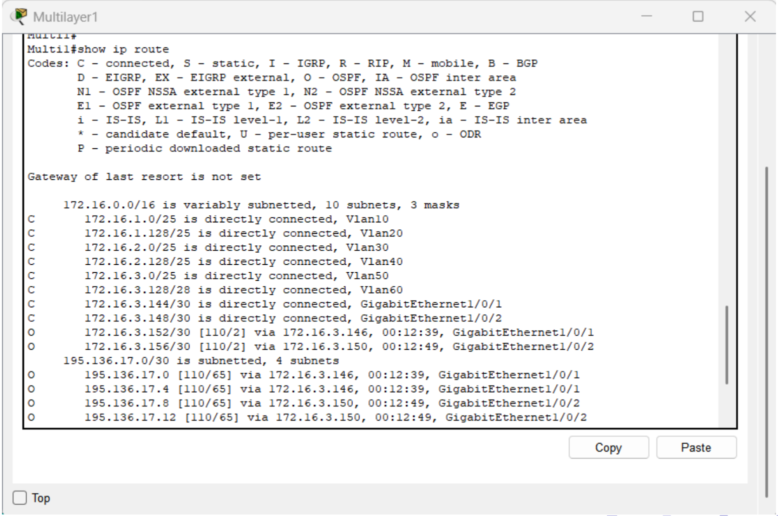


Figure: VLAN Database Verification

Dynamic Redundancy: OSPF Area 0 configured on Core Routers and Distribution Switches.

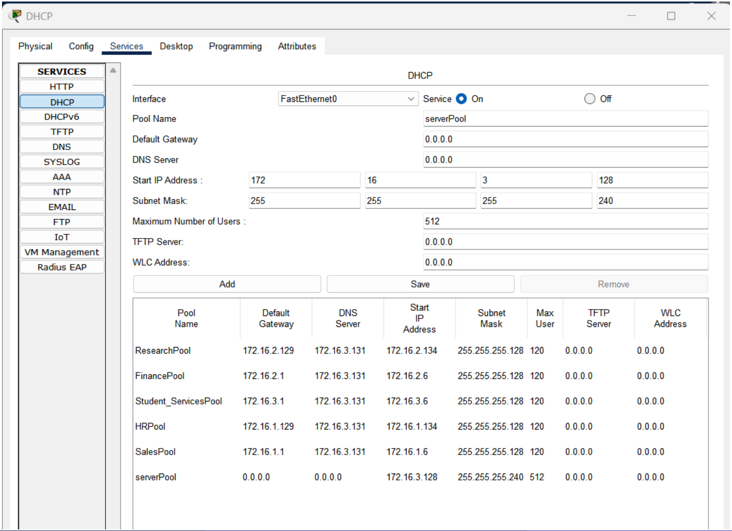



```
Multil#
Multil#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     172.16.0.0/16 is variably subnetted, 10 subnets, 3 masks
C       172.16.1.0/25 is directly connected, Vlan10
C       172.16.1.128/25 is directly connected, Vlan20
C       172.16.2.0/25 is directly connected, Vlan30
C       172.16.2.128/25 is directly connected, Vlan40
C       172.16.3.0/25 is directly connected, Vlan50
C       172.16.3.128/28 is directly connected, Vlan60
C       172.16.3.144/30 is directly connected, GigabitEthernet1/0/1
C       172.16.3.148/30 is directly connected, GigabitEthernet1/0/2
O       172.16.3.152/30 [110/2] via 172.16.3.146, 00:12:39, GigabitEthernet1/0/1
O       172.16.3.156/30 [110/2] via 172.16.3.150, 00:12:49, GigabitEthernet1/0/2
     195.136.17.0/30 is subnetted, 4 subnets
O       195.136.17.0 [110/65] via 172.16.3.146, 00:12:39, GigabitEthernet1/0/1
O       195.136.17.4 [110/65] via 172.16.3.146, 00:12:39, GigabitEthernet1/0/1
O       195.136.17.8 [110/65] via 172.16.3.150, 00:12:49, GigabitEthernet1/0/2
O       195.136.17.12 [110/65] via 172.16.3.150, 00:12:49, GigabitEthernet1/0/2
```

Automated Addressing: Centralized DHCP Server located in VLAN 60 (Server Room).



Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User	TFTP Server	WLC Address
ResearchPool	172.16.2.129	172.16.3.131	172.16.2.134	255.255.255.128	120	0.0.0.0	0.0.0.0
FinancePool	172.16.2.1	172.16.3.131	172.16.2.6	255.255.255.128	120	0.0.0.0	0.0.0.0
Student_ServicesPool	172.16.3.1	172.16.3.131	172.16.3.6	255.255.255.128	120	0.0.0.0	0.0.0.0
HRPool	172.16.1.129	172.16.3.131	172.16.1.134	255.255.255.128	120	0.0.0.0	0.0.0.0
SalesPool	172.16.1.1	172.16.3.131	172.16.1.6	255.255.255.128	120	0.0.0.0	0.0.0.0
serverPool	0.0.0.0	0.0.0.0	172.16.3.128	255.255.255.240	512	0.0.0.0	0.0.0.0

Management: SNMP Implementation

Proactive Monitoring: SNMPv2c deployed for remote device monitoring.

Configuration:

- Community: `public` (RO)
- Manager: Admin PC
- Target: Core/Distribution Devices

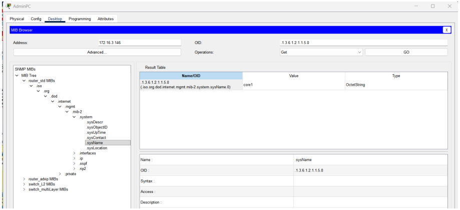


Figure: MIB Browser Result

Project Goals Achieved

- **Scalability:** Hierarchical design supports easy expansion.
- **Redundancy:** Validated OSPF neighbor adjacencies.
- **Connectivity:** Full inter-VLAN communication verified.
- **Management:** Successful SNMP query validation.

Thank You!